

E-COMMERCE RETURN RATE REDUCTION ANALYSIS

A Data Analytics & Machine Learning Case Study

Diagnosing Product Return Drivers and Enabling Data-Driven Retention Strategy

Python | SQL (MySQL) | Power BI |

Portfolio Project Report
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1. Executive Summary

Product returns represent one of the most significant hidden costs in e-commerce operations, eroding margins, straining logistics networks, and undermining customer trust. This report presents an end-to-end data analytics initiative undertaken to diagnose the root causes of product returns and to equip business stakeholders with the tools needed to reduce them.

Using a dataset of 10,000 customer orders, the analysis combined Python-based data cleaning and exploratory analysis, SQL-driven business querying, a Logistic Regression model for return-risk prediction, and an interactive Power BI dashboard for ongoing monitoring. The result is a complete, decision-ready analytics solution rather than a one-off report.

The analysis revealed a return rate of 50.5% across the sampled orders, with clothing emerging as the highest-risk product category and discounting behavior showing a measurable relationship with return likelihood. A recurring data-quality gap — nearly half of all returns lacking a documented reason — was itself identified as a priority finding, as it currently limits the organization's ability to target root causes with precision.

Taken together, the findings and the accompanying dashboard give category managers, pricing teams, and operations leaders a repeatable framework for monitoring return performance, prioritizing high-risk SKUs, and testing interventions such as revised discount thresholds and improved return-reason capture at the point of return.

2. Business Problem

Returns are a normal part of e-commerce, but excessive or poorly understood return volume creates compounding costs: reverse logistics and restocking fees, inventory write-downs, refund processing overhead, and reduced customer lifetime value when the return experience is poor.

The retailer in this analysis lacked a consolidated, quantitative view of its return performance. Return data existed, but it was not being systematically queried, segmented, or modeled to answer the questions that matter most to the business:

- Which product categories, suppliers, and regions generate a disproportionate share of returns?
- Does discounting behavior increase the likelihood that an order will be returned?
- Can orders likely to be returned be flagged before fulfillment, enabling proactive intervention?
- Where are the gaps in current return-reason data collection, and how are they limiting root-cause analysis?

This project was scoped to close that gap: to convert raw, unstructured order and return records into a governed, queryable dataset, and to layer descriptive, diagnostic, and predictive analytics on top of it so the business can act rather than merely observe.

3. Project Objectives

- Clean and standardize raw e-commerce order and return data to a production-quality analytical dataset.
- Quantify return performance at the overall, category, payment-method, and shipping-method level using SQL.
- Identify statistically meaningful patterns and correlations behind product returns using exploratory data analysis.
- Build a Logistic Regression model to estimate the probability that a given order will be returned.
- Design an interactive Power BI dashboard that enables non-technical stakeholders to self-serve return insights.
- Translate analytical findings into specific, actionable business recommendations for reducing return rates.

4. Dataset Overview

The analysis is based on a structured e-commerce transactions dataset of 10,000 customer orders, spanning order, product, customer, payment, shipping, and return attributes. The table below summarizes the core performance indicators established during the initial data profiling stage and later surfaced as headline KPIs on the Power BI dashboard.

10,000 Total Orders	5,052 Total Returns	50.5% Return Rate	24.99% Avg. Discount	\$762.62 Avg. Order Value
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4.1 Key Data Fields

- Order details: Order ID, order date, order value, discount percentage
- Product details: product category, supplier
- Customer details: geography / region, marketing channel
- Transaction details: payment method, shipping method
- Return details: return flag (Returned), return reason

5. Technology Stack

The project was intentionally built using an industry-standard, end-to-end analytics stack — spanning scripting, database querying, machine learning, and business intelligence — to mirror the toolset used by data analytics teams in production e-commerce environments.

Tool	Purpose
Python	Data cleaning, preprocessing, exploratory data analysis, and machine learning
Pandas	Data manipulation, transformation, and aggregation
NumPy	Numerical computation supporting feature engineering
Matplotlib	Static data visualization for trend and distribution analysis
Seaborn	Statistical visualization and correlation analysis
Scikit-learn	Logistic Regression model development for return prediction
SQL (MySQL)	Data storage, querying, aggregation, and business-level analysis
Power BI Desktop	Interactive dashboard design and stakeholder-facing visualization
Jupyter Notebook	Iterative Python development, documentation, and analysis

6. Methodology

The project followed a structured analytics lifecycle, moving from raw data to a validated, business-ready decision-support tool:

- Data Collection — sourcing order, customer, product, payment, shipping, and return records.
- Data Cleaning — removing duplicates, handling missing values, correcting data types, and deriving the Returned indicator.
- SQL Analysis — querying and aggregating the cleaned data to quantify return performance across key business dimensions.
- Exploratory Data Analysis — visualizing distributions, trends, and correlations to surface return-driving factors.
- Predictive Modeling — training a Logistic Regression classifier to estimate return probability at the order level.
- Dashboard Development — building an interactive Power BI report for ongoing, self-service monitoring.
- Insight Generation — translating statistical findings into prioritized, actionable business recommendations.

7. Data Cleaning Process

Data quality is the foundation of reliable analytics. Using Python and Pandas, the raw dataset was systematically audited and cleaned before any analysis was performed, with the following steps applied:

- Removed duplicate order records to prevent inflated order and return counts.
- Identified and handled missing values across product, payment, and return-reason fields using appropriate imputation or exclusion rules.
- Standardized data types (dates, numeric fields, categorical labels) to ensure consistency across downstream SQL and BI layers.
- Engineered the binary Returned indicator from raw return status fields to support both SQL aggregation and machine learning.
- Validated the final dataset for referential consistency prior to loading it into MySQL and Power BI.

This step ensured that all downstream metrics — from the dashboard KPIs to the predictive model — were built on a single, trustworthy source of truth.

8. SQL Analysis

With the cleaned dataset loaded into MySQL, structured queries were developed to quantify return performance across the dimensions most relevant to business decision-making:

- Aggregate order and return counts to establish baseline volume and overall return rate.
- Return rate by product category, to identify categories driving disproportionate return volume.
- Return rate by payment method and shipping method, to test whether transaction mechanics influence return behavior.
- Product-level summaries, to surface individual SKUs with elevated return frequency.

These queries formed the analytical backbone later surfaced through the Power BI dashboard, ensuring that every visual on the dashboard is traceable back to a governed SQL calculation.

9. Exploratory Data Analysis (EDA)

Exploratory analysis was performed in Python using Matplotlib and Seaborn to move beyond aggregate return rates and understand the underlying distributions and relationships in the data. Key analyses included:

- Return percentage by product category, to quantify category-level risk.
- Return percentage by supplier, to assess whether return risk is concentrated among specific vendors.
- Return percentage by geography, to detect regional patterns in return behavior.
- Marketing channel analysis, to evaluate whether acquisition source correlates with return likelihood.
- Correlation heatmap, to identify relationships between numeric variables such as discount percentage and order value.

- Distribution plots, to understand the spread of order values, discounts, and return frequency.
- Return trend analysis over time, to detect seasonality or emerging shifts in return behavior.

Together, these analyses moved the project from “what happened” to “why it happened” — directly informing which variables were carried forward into the predictive model.

10. Machine Learning Model

To move from historical reporting to forward-looking risk management, a Logistic Regression model was developed using Scikit-learn to estimate the probability that a given order would be returned.

10.1 Modeling Approach

- Encoded categorical variables (product category, payment method, shipping method, marketing channel) into a model-ready numeric format.
- Split the dataset into training and testing sets to enable unbiased performance evaluation.
- Trained a Logistic Regression classifier to predict return probability at the order level.
- Applied the trained model to flag high-risk products and orders for proactive review.

10.2 Business Value

Logistic Regression was selected for its interpretability: beyond producing a return-probability score, it allows the business to understand the direction and relative strength of each factor's influence on return likelihood — a critical requirement for a model that will inform pricing and merchandising decisions, not just a black-box prediction.

11. Power BI Dashboard

The final deliverable is an interactive Power BI dashboard designed for business users rather than technical analysts. It consolidates the SQL and EDA findings into a single, filterable interface that lets managers explore return performance without writing a query.

Component	Details
KPI Cards	Total Orders, Total Returns, Return Rate, Average Discount, and Average Order Value at a glance
Bar Chart	Return percentage by product category
Line Chart	Return volume trend over time
Donut Chart	Breakdown of documented return reasons
Column Chart	Return rate by payment method and shipping method
Line Charts	Impact of discount level on return rate

Detail Table	Product category, shipping method, total orders, and total returns
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The dashboard is built to support day-to-day decision-making: category managers can filter to their own product lines, finance can monitor the return rate trend against targets, and operations can drill into shipping and payment method performance — all from the same governed dataset.

12. Key Business Insights

- **Clothing is the highest-risk category.** Clothing products recorded the highest return percentage of any product category, indicating that sizing, fit, or product-description accuracy is a priority area for review.
- **Return-reason data has a significant tracking gap.** Nearly half of all returns were logged as "Not Mentioned," meaning the organization currently lacks the data needed to distinguish between quality issues, fulfillment errors, and customer preference changes at scale.
- **Discounting is correlated with return behavior.** Return rates varied with discount level across payment methods, suggesting that aggressive discounting may be attracting lower-commitment purchases that are more likely to be returned.
- **Shipping method has a limited effect on returns.** Differences in return rate across shipping methods were minor, indicating that delivery method is not a primary driver of return behavior in this dataset.
- **Predictive scoring enables proactive intervention.** The Logistic Regression model successfully identifies products with an elevated probability of return, giving merchandising and quality teams a ranked list to prioritize before returns occur rather than after.

13. Challenges Faced

- **Incomplete return-reason data:** a substantial share of return records lacked a specific reason code, which limited the granularity of root-cause analysis and required the team to treat this gap itself as a reportable finding.
- **Categorical variable encoding:** product, payment, and shipping fields required careful encoding to preserve their business meaning while remaining usable in the Logistic Regression model.
- **Balancing model interpretability with performance:** Logistic Regression was chosen over more complex algorithms specifically to keep the model's reasoning transparent to non-technical business stakeholders.
- **Aligning SQL, Python, and Power BI outputs:** ensuring that KPI definitions (e.g., return rate calculation) were identical across the SQL layer, the Python EDA, and the dashboard required deliberate consistency checks.

14. Future Improvements

- Enforce mandatory, structured return-reason capture at the point of return to close the current data-quality gap and enable deeper root-cause analysis.
- Extend the predictive model with additional algorithms (e.g., Random Forest, Gradient Boosting) and compare performance against the current Logistic Regression baseline.
- Incorporate customer-level features, such as purchase history and prior return behavior, to improve prediction accuracy.
- Automate the data pipeline (Python → MySQL → Power BI) using scheduled refreshes to support near real-time monitoring.
- Add a financial-impact layer to the dashboard, translating return volume into estimated cost and margin impact for leadership reporting.

15. Assumptions & Limitations

- The dataset represents a defined historical sample of 10,000 orders and may not capture all seasonal or category-specific variation present in the full production dataset.
- Return reasons marked "Not Mentioned" were treated as a distinct, uncategorized group rather than imputed to a specific cause, to avoid introducing unsupported assumptions into the analysis.
- The Logistic Regression model assumes a linear relationship between encoded features and the log-odds of return; more complex, non-linear interactions may not be fully captured.
- Findings reflect correlation, not confirmed causation; recommendations (e.g., around discounting) should be validated through controlled testing before broad policy changes are made.

16. Conclusion

This project demonstrates a complete, production-style analytics workflow — from raw data to governed dataset, from SQL-based diagnostics to predictive modeling, and from static findings to a live, interactive dashboard. It reflects the core skill set expected of a modern data analyst: the ability to clean and structure data, query it rigorously, model it responsibly, and communicate it in a way business stakeholders can act on.

The analysis identified clothing as the highest-risk category, surfaced a material gap in return-reason data collection, and established a measurable link between discounting behavior and return likelihood. The accompanying predictive model and Power BI dashboard turn these findings into an operational tool, enabling the business to monitor return performance continuously and intervene on high-risk products before returns occur.

More broadly, this project illustrates how integrating Python, SQL, and Power BI into a single workflow can convert raw transactional data into a repeatable, decision-ready business intelligence solution — one that supports better inventory management, more disciplined pricing strategy, improved product quality control, and a stronger overall customer experience.